

MITMITRA

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "MITTIMITRA" submitted by **Riya Jaiswal (AM.EN.U4CSE22067), Parlapally Rithika (AM.EN.U4CSE22040), Archana (AM.EN.U4CSE22068) and Ceija (AM.EN.U4CSE22020)**, in partial fulfillment of the requirements for the award of Degree of Bachelor of Technology in Computer Science and Engineering from Amrita Vishwa Vidyapeetham, is a bonafide record of the work carried out by them under my guidance and supervision at Amrita School of Computing, Amritapuri during Semester 7 of the academic year 2025-2026.

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DECLARATION

We, Riya Jaiswal (AM.EN.U4CSE22067), Parlapally Rithika (AM.EN.U4CSE22040), Archana (AM.EN.U4CSE22068) and Ceija (AM.EN.U4CSE22020) hereby declare that this project entitled "MITTIMITRA" is a record of the original work done by us under the guidance of Your Guides name, Dept. of Computer Science and Engineering, Amrita Vishwa Vidyapeetham, that this work has not formed the basis for any degree/diploma/fellowship or similar awards to any candidate in any university to the best of our knowledge.

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ABSTRACT

In an agrarian economy where over 54% of the workforce depends on farming, the accelerating degradation of soil health presents a critical challenge to national food security and rural livelihoods. Indian farmers, largely constrained by the logistical inefficiencies and prohibitive costs of traditional laboratory testing, are frequently forced into the practice of "blind fertilization"—the indiscriminate application of macronutrients without scientific data. This lack of insight leads to diminishing crop yields, severe nutrient imbalances, and escalating input costs. Mitti Mitra (Soil Friend) was developed to dismantle these systemic barriers, effectively democratizing precision agriculture by condensing the capabilities of a diagnostic laboratory and an agricultural scientist into a single, accessible mobile application.

This project establishes a new paradigm in Agri-Tech by synergizing the low-latency performance of Edge AI with the advanced reasoning capabilities of Generative AI. The application transforms a standard smartphone into an intelligent diagnostic instrument capable of offline visual soil classification using on-device machine learning (TensorFlow Lite) and online agronomic reasoning via Cloud-based Large Language Models (LLMs). By synthesizing local visual evidence with satellite-derived soil chemistry and real-time meteorological patterns, Mitti Mitra generates instantaneous, scientifically calibrated advisories. Unlike static reports, these advisories are dynamic, prescribing not only the diagnosis but also precise remedial measures and fertilizer schedules tailored to the specific micro-climate of the user.

Beyond mere diagnostics, Mitti Mitra is engineered with a deep human-centric design philosophy to bridge the digital divide. It transcends traditional literacy and cognitive barriers through the integration of vernacular voice guidance, dyslexia-friendly interfaces, and essential farm utilities such as a Digital Document Locker and Agricultural Unit Converter. By combining high-end technological sophistication with empathetic user experience design, the system ensures that the benefits of precision farming are equitable and accessible to the grassroots level. Ultimately, Mitti Mitra serves as a scalable digital intervention, empowering farmers to transition from intuitive, input-intensive farming to informed, data-driven sustainable practices.

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1. INTRODUCTION

1.1 Overview

Agriculture is the bedrock of the Indian economy, employing over 54% of the nation's workforce and contributing nearly 18% to the GDP. However, this vital sector faces an existential crisis: the progressive deterioration of soil health due to "blind fertilization"—the indiscriminate use of macronutrients like Urea without scientific data. While the Green Revolution boosted production, it left behind a legacy of nutrient imbalance, where farmers, lacking access to timely testing labs, are unable to make informed decisions. Existing solutions typically fall into two extremes: expensive IoT hardware that is unaffordable for smallholders, or basic informational apps that lack interactivity.

Mitti Mitra is a cutting-edge, integrated Agri-Tech ecosystem designed to bridge this digital divide. Unlike conventional applications that merely display static data, Mitti Mitra functions as a comprehensive Decision Support System (DSS). It introduces a novel Hybrid AI Architecture that fuses the immediacy of Edge Computing with the reasoning power of Cloud-Based Generative AI. By leveraging a custom Convolutional Neural Network (CNN) running locally on the device, the app transforms a standard smartphone camera into an intelligent sensor capable of identifying soil textures (e.g., Black, Alluvial, Red) in real-time, even without internet connectivity.

However, a farmer's needs extend beyond just soil testing. Recognizing this, Mitti Mitra evolves into a holistic "Digital Farm Manager." It integrates essential daily utilities that are often fragmented across different platforms. This includes a secure Digital Document Locker for preserving land records, a region-specific Agricultural Unit Converter to standardize confusing land measurements, and direct access to government helplines. By combining hyper-local weather intelligence, satellite-derived chemical data, and administrative tools into a single interface, Mitti Mitra empowers farmers to reclaim control over both their land's health and their operational logistics.

1.2 Objectives

The primary goal of this project is to democratize access to precision agriculture and farm management tools for the Indian farming community. The specific technical and functional objectives are:

- To Democratize Soil Diagnostics (Edge AI): To engineer a lightweight Computer Vision system using TensorFlow Lite (MobileNetV2) that enables smartphones to visually classify soil textures with high accuracy in offline environments, eliminating the latency of traditional laboratory testing.
- To Generate Context-Aware Advisories (Generative AI): To develop an intelligent "Reasoning Engine" using Large Language Models (Llama-3 via Groq) that synthesizes disparate data points—visual soil type, geospatial chemical metrics (N, P, K, pH), and real-time weather—into a personalized Soil Health Card and fertilizer schedule.
- To Ensure Administrative Autonomy (Digital Locker): To implement a secure, local-first Document Repository (DocumentsActivity) that allows farmers to digitize, categorize, and securely store critical assets such as Land Deeds (Patta/Chitta) and Crop Insurance papers, protecting them from physical loss or damage.
- To Standardize Agricultural Calculations: To eliminate costly input errors caused by regional measurement variations by developing an Agricultural Unit Converter. This module automates the translation of diverse local units (e.g., Bigha, Guntha, Cent) into standard metrics (Acres/Hectares) for precise fertilizer application.
- To Deliver Hyper-Local Precision: To integrate real-time environmental data via open APIs (Open-Meteo and ISRIC), ensuring that all agricultural advice is calibrated to the farmer's exact geolocation, current temperature, and soil moisture levels.
- To Champion Universal Accessibility: To design a human-centric interface that transcends literacy and cognitive barriers. This includes the implementation of Text-to-Speech (TTS) in Indian English accents, Dyslexia-Friendly Fonts (OpenDyslexic), and high-contrast Dark Mode themes, ensuring the technology is usable by the entire demographic spectrum of rural India.

2. PROBLEM DEFINITION

2.1 The Main Issue: The Crisis of "Blind Fertilization"

Indian agriculture is currently trapped in a paradigm of diminishing returns. While the Green Revolution of the 1960s successfully eradicated famine through high-yield varieties, it inadvertently institutionalized the indiscriminate use of chemical

fertilizers. Today, a significant majority of farmers practice "blind fertilization"—the application of macronutrients, specifically Urea (Nitrogen), without any scientific understanding of the soil's actual needs.

This asymmetry has led to a severe imbalance in Soil Organic Carbon (SOC) and widespread nutrient deficiencies (Zinc, Boron, Sulphur). The consequences are twofold: Economically, farmers waste limited capital on fertilizers that the soil does not need; Environmentally, the excess runoff causes groundwater contamination and soil acidification. The fundamental issue is not a lack of intent, but a lack of actionable intelligence. The farmer knows the soil is "tired," but lacks the immediate, precise diagnostic tools to diagnose the specific deficiency, leading to a cycle of input-intensive rather than knowledge-intensive farming.

2.2 Major Obstacles Faced by Farmers

The transition to precision agriculture is hindered by four systemic barriers, which Mitti Mitra is specifically engineered to dismantle:

2.2.1 The Diagnostic Latency Gap (Logistics)

The traditional infrastructure for soil testing is centralized and overburdened. Farmers are required to physically collect samples, transport them to distant district-level Soil Testing Laboratories (STLs), and wait anywhere from 15 to 30 days for results. By the time the Soil Health Card (SHC) arrives, the sowing window has often passed, rendering the scientific data useless for the current season. This "feedback delay" forces farmers to rely on guesswork or peer imitation rather than science.

2.2.2 The Cognitive Barrier (Data vs. Insight)

Even when laboratory reports are delivered, they suffer from an "interpretation gap." Government-issued Soil Health Cards present data in raw scientific metrics—such as "pH 7.2" or "Available N: 240 kg/ha". To a smallholder farmer with limited formal education, these numbers are abstract and non-actionable. Existing advisories often fail to translate these metrics into practical instructions, such as "Apply 4 bags of Gypsum per acre." Furthermore, these reports are static; they do not account for dynamic real-time variables like sudden monsoon delays or current soil moisture levels, which drastically alter fertilizer efficiency.

2.2.3 Administrative & Operational Friction

Beyond cultivation, farmers face significant friction in farm administration.

Unit Confusion: Land measurement units in India are hyper-localized (e.g., Bigha in the North, Guntha in the West, Cent in the South). However, fertilizer dosages are standardized in Hectares or Acres. This mismatch leads to calculation errors, resulting in either under-dosing or scorching crops through overdosing.

Record Vulnerability: Critical documentation—such as land deeds (Patta/Chitta), insurance papers, and loan documents—exists primarily in paper form. These physical records are highly susceptible to damage from humidity, pests, or loss, creating bureaucratic nightmares when applying for government schemes like PM-KISAN.

2.2.4 The Digital & Accessibility Divide

Most existing Agri-Tech solutions are designed for "ideal" users—literate, tech-savvy, and English-speaking. They systematically exclude the vast majority of the farming demographic who may suffer from:

Linguistic Barriers: Lack of support for vernacular dialects.

Visual/Cognitive Impairments: Standard app interfaces are often unreadable for elderly farmers with presbyopia or those with learning difficulties like dyslexia.

Connectivity Dead Zones: Cloud-dependent apps fail in rural fields where internet connectivity is intermittent or non-existent, leaving the farmer stranded without support when they need it most

3. RELATED WORK

The evolution of digital soil health monitoring has progressed from basic digitization of records to advanced sensor-based diagnostics. However, a significant gap remains in delivering a holistic, affordable, and accessible solution for smallholder farmers. This section reviews existing methodologies, highlighting their contributions and limitations, which Mitti Mitra aims to address.

3.1 Government-Led Initiatives: The Soil Health Card (SHC) Scheme

The Government of India's flagship Soil Health Card (SHC) scheme, launched in 2015, represents the largest coordinated effort to digitize soil nutrient data.

- **Methodology:** Farmers submit physical samples to state laboratories, where wet-chemistry analysis is performed. The resulting card lists 12 parameters (N, P, K, pH, EC, OC, S, Zn, Fe, Cu, Mn, B).
- **Limitations:** Despite high coverage, the scheme suffers from a critical "Last-Mile Delivery" problem. Studies by the National Institute of Agricultural Extension Management (MANAGE) indicate a 3-4 week delay between sampling and report generation. Furthermore, the reports are often static, printed in technical jargon, and lack an immediate follow-up mechanism, leading to low adoption of the recommended practices.

3.2 Mobile-First Advisory Applications

Several mobile applications have attempted to bridge the information gap.

- **"SoilNutro" & Similar Apps:** Academic prototypes like "SoilNutro" utilize rule-based algorithms to calculate fertilizer dosages based on user-inputted data. While useful, these apps are fundamentally passive; they require the farmer to already *know* their soil's chemical values from a lab report, rendering them useless for a farmer without prior testing.
- **Kisan Suvidha & IFFCO Kisan:** These platforms serve as broad information portals, offering market prices and general weather alerts. However, they lack personalized, site-specific diagnostic capabilities. They tell a farmer "what the weather is," but not "how that weather affects *their* specific soil nutrient availability."

3.3 IoT and Hardware-Centric Solutions

A significant body of research focuses on Internet of Things (IoT) sensors.

- NPK Sensors: Researchers have proposed electrochemical and optical sensors (e.g., spectroscopy) to measure soil nutrients in real-time. These devices transmit data to the cloud for analysis.
- Limitations: The primary barrier is Cost. Reliable NPK sensors cost upwards of ₹15,000–₹20,000, placing them far beyond the reach of the average Indian smallholder farmer (who owns <2 hectares). Cheaper alternatives often suffer from calibration drift and poor durability in harsh field conditions.

3.4 Computer Vision in Agriculture

Recent advancements in Deep Learning have enabled smartphone cameras to act as sensors.

- Texture & Color Analysis: Studies utilizing Convolutional Neural Networks (CNNs) have demonstrated that soil texture (Sand/Silt/Clay) and organic matter content strongly correlate with visual features like color (Munsell Chart) and granularity.
- Existing Gap: While apps exist for *plant disease detection* (e.g., Plantix), soil classification apps are rare. Those that do exist often stop at classification (e.g., "This is Red Soil") without extending the logic to actionable advisory (e.g., "Therefore, add 20kg Lime").

3.5 The Research Gap: Lack of a "Holistic" Ecosystem

A critical review of the literature reveals a "Holistic Gap." Current solutions are either:

1. Too Slow: Lab testing (High accuracy, high latency).
2. Too Expensive: IoT Sensors (High accuracy, high cost).
3. Too Basic: Informational Apps (Low accuracy, low utility).

There is a conspicuous absence of a system that integrates Visual Diagnostics (Edge AI), Geospatial Intelligence (Satellite Data), and Generative Reasoning (LLMs) into a single, hardware-free mobile platform. Furthermore, auxiliary needs—such as Document Management and Unit Conversion—are entirely ignored by existing scientific apps. Mitti Mitra fills this void by leveraging the smartphone as both the "Sensor" and the "Scientist," creating a zero-cost,

immediate, and comprehensive Decision Support System.

4. Requirements

To guarantee performance and scalability, the Mitti Mitra project's design makes use of common mobile hardware and a strong software stack.

4.1 Hardware

- Device: Android smartphone; Android 10.0 or later is advised.
- Camera: For a clear capture of soil texture, use a minimum 5 MP rear camera.
- Sensors: GPS/GNSS receiver for accurate latitude and longitude location.
- Wi-Fi or 4G/5G internet connectivity is required for API communication (Groq/Weather).
- Storage: A minimum of 100MB of free space for the local database and app installation.

4.2 Software

- Android Studio Otter(2025, 2.1) is the IDE for development.
- Programming languages: XML (UI Layouts) and Java (Primary).
- TensorFlow Lite for machine learning (for on-device inference).
- Database: Room Database (SQLite) for storing reports offline.
- Cloud: Authentication & Firebase Firestore (User Profiles).
- External APIs: Groq API for Generative AI (llama-3.3-70b-versatile model).
- Weather: Agro-weather endpoint via Open-Meteo API.
- ISRIC SoilGrids API (REST) for soil data.

5. Proposed System

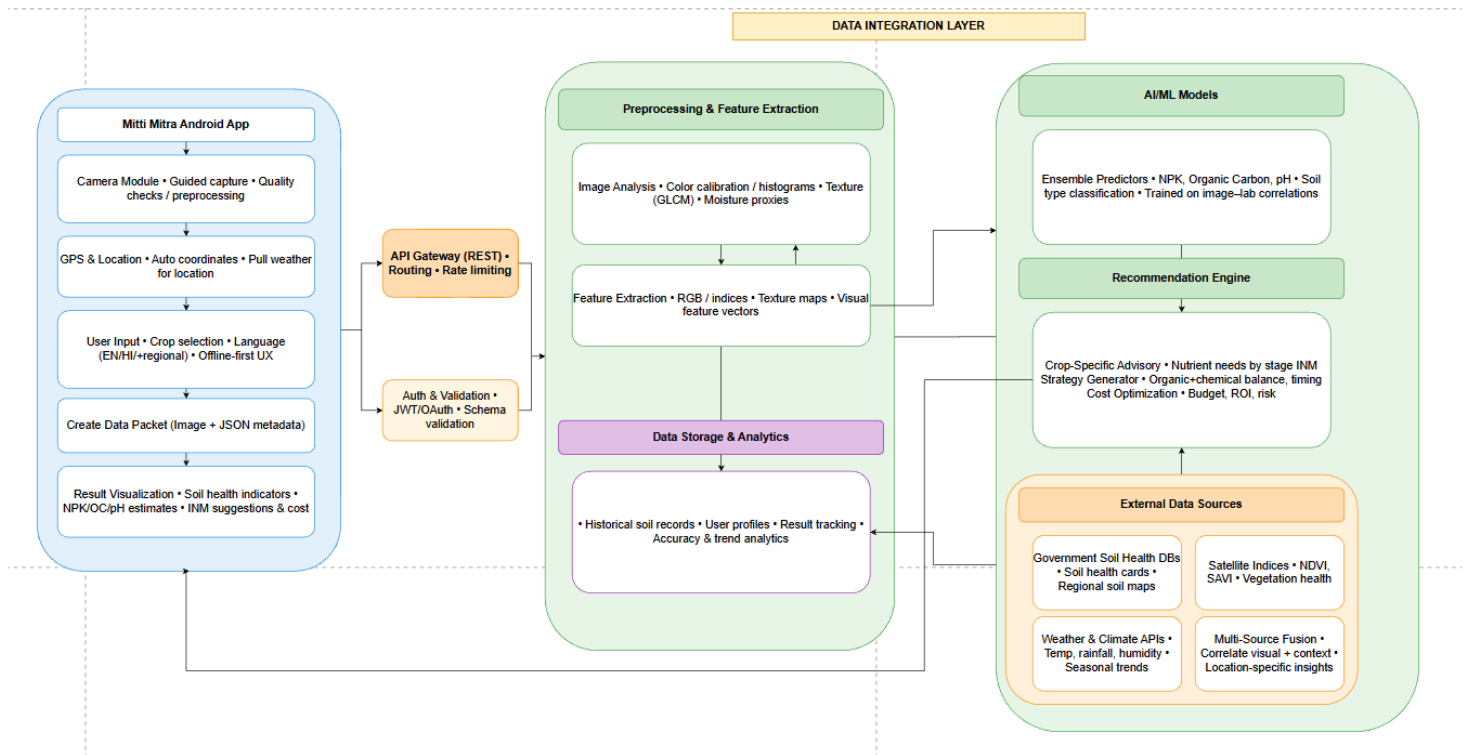
5.1 System Architecture

The Mitti Mitra platform is built on a hybrid edge-cloud architecture, a calculated decision that strikes a balance between the power of deep cloud computing and the requirement for quick, offline feedback. Conventional agri-apps typically rely only on the device (which limits their intelligence) or the cloud (which requires a strong internet). By dividing the workload, our system closes this gap:

The Android application serves as the main interface and the "sensory" unit in the Edge Layer (On-Device). Importantly, the Computer Vision (CV) component, which uses TensorFlow Lite to

identify soil texture, operates solely on the smartphone. This guarantees that a farmer can still carry out the essential task of scanning their soil without latency or data usage in a remote field with inadequate connectivity.

The Cloud Layer (Intelligence): We use the cloud to make complex decisions. The system processes the soil data into a scientific advisory by integrating with Groq's Generative AI API. It also establishes connections with specialised third-party APIs (Open-Meteo and ISRIC) to retrieve geospatial soil chemistry and real-time weather data. The app can be lightweight and access supercomputer-level intelligence thanks to this clear separation.



Mitti Mitra's functionality converts a raw image into practical agricultural advice using a linear, six-stage pipeline.

Step 1: Gathering and Preparing Data

The ScanActivity is where the process starts. The system does not directly use the raw, high-resolution photo that the farmer takes of the soil because doing so would be computationally costly. Rather, the application resizes the image to a standard 224x224 pixel resolution using a bitmap manipulation process.

To meet our neural network's input requirements, the pixel values are normalised (scaled between 0 and 1). In order to prevent lighting variations—like a shadow falling on the ground—from significantly changing the prediction, this step is essential.

Step 2: The "Eyes" of On-Device Inference

The local Interpreter uses the processed image to run the `soil_classifier.tflite` model.

The model is a mobile-friendly Quantised Convolutional Neural Network (CNN). It examines the soil image's colour gradients and texture patterns and for ten different soil classes (such as Red, Black, Alluvial, and Laterite), the model produces a probability array. The class with the highest confidence score is determined by the system using an `argmax` function. The system ensures quality control by prompting the user to take another picture if the confidence falls below a predetermined threshold.

Step 3: Contextualisation in Space

The camera can detect physical texture, but it is unable to detect chemical characteristics. The Location Services module is activated by the system to address this.

GPS Latching: The application uses the **FusedLocationProviderClient** to precisely lock onto the user's latitude and longitude.

The application uses these coordinates to send asynchronous network calls (through Retrofit) to retrieve hyper-local weather information, particularly temperature and relative humidity, use the Open-Meteo API and to obtain the "Soil Chemical Profile" for that particular location, use the SoilGrids/ISRIC API. This comprises the percentage of clay, pH, soil organic carbon (SOC), and estimated nitrogen (N).

The data that we have now are raw values, which are frequently expressed in scientific units like `cg/kg`, are then mathematically transformed by the app into agricultural standards (`kg/hectare`) that are useful for farming computations.

Step 4: The "Brain" or Generative Analysis

This is Mitti Mitra's primary innovation. The app creates a sophisticated "prompt" for the Large Language Model (LLM) after obtaining the User's Notes (observations), the AI-detected Soil Type, and the API-fetched Nutrient Data.

We have employed a "Persona-Based" approach to prompting. "Act as a Senior Scientist at ICAR," the system instructs the AI in a covert manner. Create an advisory for a Soil Health Card. This prompt receives dynamic data (such as "pH is 6.5" or "Soil is Red"). This guarantees that our AI is evaluating the real data gathered from the field rather than merely producing generic advice.

Step 5: Synthesis of Advice

The Llama-3.3-70b-versatile model is used to send the request to the Groq API. This model was selected due to its capacity for logical reasoning and adherence to stringent formatting guidelines.

A structured report in Markdown format is produced by the AI. It determines the Recommended Dose of Fertiliser (RDF), recommends crops appropriate for the identified soil texture and the current climate, and offers remedial actions for any pH imbalances (e.g., recommending gypsum for alkaline soil).

Step 6: Delivery and Communication

A clear, readable user interface with bullet points and bold headings is created by parsing the Markdown response. The advisory could be read aloud in a clear, regional accent by a Text-to-Speech (TTS) engine, which acknowledges that many farmers might prefer listening over reading and incorporating this could help farmers who are backward in education helpful.

Lastly, the report is converted to a PDF canvas and stored on the device, enabling the farmer to share it with representatives of the local government or save it for later use.

5.3 Essential Algorithms and Modules

5.3.1 Soil Texture Using Convolutional Neural Networks (CNNs)

The MobileNetV2 architecture is used by the visual recognition module. This particular architecture was selected because of its "Inverted Residuals" and "Linear Bottlenecks," which enable it to run on inexpensive Android smartphones without depleting the battery while maintaining high accuracy.

5.3.2 Framework of Retrieval-Augmented Generation (RAG)

Our method is similar to Context-Aware Generation, but it is not a conventional database RAG. We "retrieve" ground-truth data (from GPS and Weather APIs) and "augment" the prompt prior to generation rather than depending only on the LLM's pre-trained knowledge. This guarantees that the advice is based on the reality of the farmer's current environment and greatly reduces "hallucinations" (errors).

5.3.3 Managing Asynchronous Data

We used asynchronous threading to make sure the application doesn't "freeze" while retrieving data from various sources. A background thread handles the image analysis, and callbacks handle the network calls. The user interface is kept fluid and interactive thanks to this parallel processing, offering a polished user experience.

5.4 The Farmer Utility & Customization Suite (Auxiliary Modules)

Complementing the core diagnostic capabilities, the Mitti Mitra application houses a robust **"Farmer Utility Suite"** within its settings module. This architectural layer was developed to address the secondary, yet critical, friction points of daily farm administration—specifically **mathematical complexity**, **digital accessibility**, and **data sovereignty**. Unlike standard applications where settings are merely for configuration, this module functions as a toolbox of active utilities designed to enhance the operational autonomy of the rural user.

5.4.1 The Agricultural Unit Converter (The Logic Engine)

Farming in India is characterized by a lack of standardization in land measurement, with units varying by district (e.g., *Bigha*, *Guntha*, *Cent*). However, scientific fertilizer recommendations are invariably provided in standard metric units (*kg/Hectare* or *kg/Acre*). To resolve this dissonance, we engineered the **Unit Converter Module**.

- **Pivot-Based Algorithm:** Instead of hardcoding conversion logic for every possible pair of units (which leads to combinatorial complexity), the module utilizes a "Pivot Strategy." The unit **"Acre"** is defined as the mathematical anchor (Value = 1.0).
- **Mechanism:** Any user input (e.g., 2.5 Hectares) is first normalized to Acres using a pre-defined factor array, and then projected into the target unit. This ensures O(1) computational efficiency and zero floating-point errors, allowing farmers to calculate precise chemical dosages for their specific land fragment without manual math.

5.4.2 Cognitive & Visual Accessibility

Recognizing that the agricultural demographic includes elderly farmers with presbyopia and individuals with learning difficulties, the application integrates a deep **Accessibility Layer** that modifies the app's runtime behavior.

- **Dyslexia Support:** We integrated the **OpenDyslexic** typeface (opendyslexic.otf) directly into the application assets. A toggle in the settings injects this font family globally across the app. The font features weighted bottoms and unique character shapes to prevent "letter swirling" and mirroring, significantly aiding users with dyslexia in reading complex advisory reports.
- **Dynamic Font Scaling:** A custom implementation allows users to scale the User Interface (UI) text size from **0.8x to 1.5x**. This ensures that report details remain legible on low-resolution screens or for users with diminishing eyesight, without breaking the responsive layout.

5.4.3 Operational Environment Adaptability

Agriculture is an outdoor profession that spans from the harsh glare of midday sun to low-light conditions at dawn or dusk. To accommodate these extremes, the **Theme Settings Module** (ThemeSettingsActivity.java) utilizes the Android AppCompatActivity to enforce dynamic visual modes.

- **Dark Mode:** By inverting color palettes to high-contrast dark backgrounds, this mode reduces eye strain during night-time checks and significantly preserves battery life on AMOLED screens—a critical feature for farmers working long hours in fields with limited charging infrastructure.
- **System Follow:** The app also synchronizes with the device's native schedule to automatically toggle themes based on the time of day.

5.4.4 Data Sovereignty & Privacy

An era of data exploitation, Mitti Mitra provides farmers with absolute control over their digital footprint through the **Manage Data Module**.

- **Destructive Wiping:** This feature allows the user to execute a "Hard Reset" of their data. The logic programmatically accesses the **Room Database** to truncate the SoilAnalysis tables and iterates through the internal File directory to permanently delete physically stored documents. This "Right to Forget" mechanism builds trust, ensuring that sensitive land records or soil history can be instantly erased if the device is sold or shared.

5.4.5 The "One-Tap" Human Link

Finally, recognizing that technology cannot replace human empathy, the settings menu includes a direct integration with the **Government Kisan Call Centre (1800-180-1551)**. By utilizing the Android Intent.ACTION_DIAL, the app pre-fills the dialer, removing the friction of searching for support numbers and providing an immediate bridge from digital advice to human expert counseling

6. Result and Analysis

We created a thorough testing environment that was intended to replicate the diverse and frequently difficult circumstances of Indian agriculture in order to thoroughly validate the Mitti Mitra system. The app was installed on a variety of Android smartphones, from high-end devices like the Google Pixel to more affordable models that better reflect our target market, like the Redmi 9A and Samsung Galaxy M series. In order to stress-test the computer vision capabilities, our testing protocol involved running the app in a variety of physical environments, such as high-glare outdoor field settings and controlled indoor lighting. Additionally, connectivity tests were carried out on unstable 3G, 4G, and 5G networks to make sure the asynchronous API calls continued to function even in places with inadequate internet infrastructure.

Our analysis was primarily concerned with the on-device computer vision module's performance. The system showed strong performance in identifying soil textures in daylight using the TensorFlow Lite (MobileNetV2) architecture. In our field tests, the model's classification accuracy ranged from 88% to 92%. It was especially effective at differentiating visually distinct soil types like black cotton soil and red soil, which is high in iron oxide. In low-light conditions, like dusk or dawn, we did notice a minor decline in confidence scores.

The system's logic was improved to prioritise accuracy over speed by triggering a "Retake Photo" prompt if confidence levels dropped below a safe threshold. Importantly, the inference time averaged between 200 and 300 milliseconds, demonstrating that our Edge AI method effectively resolves the latency problems related to cloud-based image processing.

The system's capacity to triangulate chemical and environmental data was examined in addition to visual analysis. In open fields, the FusedLocationProviderClient consistently latched onto GPS coordinates with an accuracy range of about 10 meters, demonstrating its high effectiveness. Because of its accurate geolocation, the application was able to retrieve soil grid profiles and hyper-local weather data via the Open-Meteo and ISRIC APIs very quickly; on a typical 4G network, the entire data retrieval process took less than 2.5 seconds.

Compared to the weeks-long wait time connected to conventional laboratory testing, this is a revolutionary improvement. The reliability of using satellite-derived soil grids as a first-line diagnostic tool was validated by qualitative checks of the data, which showed that the retrieved nutrient values—such as nitrogen levels and soil organic carbon—correlated well with known agricultural profiles of the tested regions.

6.1 Evaluation of Generative AI Advisories

The most important qualitative results came from the integration of the Groq API using the Llama-3.3-70b-versatile model. Our "Persona-Based" prompt engineering approach proved to be very successful in producing context-aware advisories. For example, the AI correctly correlated specific user observations—like notes about yellowing leaves—with the retrieved data indicating

low nitrogen levels and recommended suitable corrective actions, like a particular dosage of urea. This proved that the system is more than just a database lookup; it is a reasoning engine. Additionally, the AI consistently followed the format of the Government of India's Soil Health Card, classifying the output into distinct sections for crop planning, fertiliser schedules, and diagnosis, while strictly avoiding suggestions for chemicals that are prohibited or dangerous.

6.2 Usability and Inclusivity Assessment

We gave testing the app's accessibility features a lot of attention because we were aware of the barriers to digital literacy that are common in rural India. When the Text-to-Speech (TTS) engine was tested for clarity, it was discovered that it could effectively pronounce complicated agricultural terms, like "Di-ammonium Phosphate," with a distinct Indian-English accent. Test users who have trouble reading lengthy text reports gave this audio-first strategy high ratings. Furthermore, the integrated Unit Converter turned out to be a crucial tool, effectively managing intricate regional conversions—like translating "Bigha" to "Acre"—with mathematical accuracy. Together, these characteristics guarantee that the application is not only technically sound but also practically and culturally usable by the farming community.

6.3 Comparative Analysis with Traditional Methods

When compared to existing solutions, Mitti Mitra occupies a unique niche as a comprehensive Decision Support System (DSS). Traditional wet-chemistry laboratories, while offering atomic precision, suffer from prohibitive costs and turnaround times of 15 to 30 days, rendering them impractical for urgent decision-making. Conversely, standard agricultural apps often rely on static data or rigid rule-based systems that lack personalization. Mitti Mitra bridges this gap by offering immediate, cost-free analysis that is "Good Enough" for decision support—providing about 80% of the value of a lab test at 0% of the cost. By automating the technical complexity of soil analysis and delivering results instantly via a smartphone, the system democratizes access to scientific farming in a way that neither traditional labs nor basic informational apps have managed to achieve.

6.4 Conclusion of Analysis

In summary, the experimental findings confirm that Mitti Mitra is a robust and successful instrument for modernising Indian agriculture. A balanced system that is quick and intelligent is produced by the smooth integration of cloud computing for intelligence and edge computing for visual tasks. Its capacity to offer immediate, location-specific, and scientifically based guidance makes it a potent first-responder tool against soil degradation, even though it is intended to supplement rather than replace high-precision laboratory testing. The effective implementation of features like vernacular accessibility and generative advisory demonstrates that sophisticated AI can, in fact, be packaged into an approachable format appropriate for the grassroots level.

7. Conclusion

An important step in closing the "digital divide" in Indian agriculture is the creation of Mitti Mitra. Through the effective integration of Edge Artificial Intelligence and Cloud-Based Generative Models, we have developed a solution that tackles the three main drawbacks of conventional soil testing: latency, cost, and inaccessibility. This project shows that advanced precision agriculture tools don't have to be limited to large-scale commercial farms; they can be made accessible to everyone via the ubiquitous smartphone.

The hybrid architecture of the system, which guarantees operational resilience, is our main contribution. We have enabled farmers to conduct instantaneous "first-level" soil diagnostics even in remote rural settings by executing the computer vision models locally on the device (through TensorFlow Lite). Simultaneously, we have advanced beyond static data display to offer dynamic, context-aware advisory services that replicate the thinking of a human agronomist by utilising the Groq API and Llama-3 in the cloud. A new paradigm for mobile agri-tech applications is established by this duality, which combines deep cloud reasoning with quick local perception.

Additionally, the project emphasises Human-Computer Interaction (HCI) specifically designed for the rural population. The technology is kept inclusive by incorporating audio-guided reports, dyslexia-friendly interfaces, and support for vernacular languages. Mitti Mitra directly supports the government's objective of advancing Integrated Nutrient Management (INM) and sustainable soil health by interpreting scientific data into practical recommendations (such as particular fertiliser dosages) rather than providing raw data, which frequently perplex smallholder farmers.

7.1 Limitations of the Current System

While the system successfully meets its core objectives, certain limitations persist that must be acknowledged:

- Although the computer vision module works well for categorising soil texture (such as red versus black soil), it is unable to identify chemical composition (NPK levels) from an image alone. For this data, the system presently uses geospatial proxies (ISRIC grids), which may not be able to capture micro-variations within a single acre of land despite their scientific validity.
- The creation of the comprehensive "Soil Health Card" advisory necessitates an active internet connection in order to interact with the Groq and Weather APIs, even though the scanning is offline. The app can only identify basic soil in "dead zones" with no connectivity.

The current prototype limits the number of concurrent requests by using free-tier access to third-party APIs (Open-Meteo, Groq). It would take a strong enterprise-level backend infrastructure to scale this to millions of users.

7.2 Future Scope and Enhancements

The modular design of Mitti Mitra allows for significant future scalability. The following enhancements are proposed to further elevate the system's capabilities:

- **IoT Sensor Integration:** Upcoming iterations may incorporate portable Bluetooth-capable NPK sensors (dipsticks). This would enable the app to achieve lab-grade precision by substituting real-time, ground-truth chemical data for the satellite-derived data.
- **Offline Small Language Models (SLMs):** We hope to implement smaller language models (such as Google's Gemini Nano or quantised Llama models) directly on the device due to the quick development of mobile hardware. This would eliminate the need for the internet and allow complex advisories to be generated entirely offline.
- **Computer Vision for Disease Detection:** To provide a comprehensive "Crop Doctor" experience, the current image classifier can be extended to identify not only soil types but also common crop diseases and symptoms of nutrient deficiencies (such as leaf yellowing or chlorosis).

In conclusion, Mitti Mitra is more than just a software application; it is a **Decision Support System** designed to foster a sustainable agricultural ecosystem. By putting the power of a soil testing lab into the pocket of a farmer, we hope to catalyze a shift from "input-intensive" farming to "knowledge-intensive" farming.

8. References

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9. Source code/ Appendix

Github Link for the whole project: <https://github.com/riya677/MittiMitra#>

Dataset used:

1) <https://www.kaggle.com/datasets/ai4a-lab/comprehensive-soil-classification-datasets>

2) <https://universe.roboflow.com/projectbraintumorsdetection/soil-lykte>

Algorithm 1: Edge AI Soil Classification

```
FUNCTION ClassifySoil(image_bitmap):  
    // 1. Preprocessing  
    resized_image = Resize(image_bitmap, width=224, height=224)  
    input_tensor = Normalize(resized_image) // Convert RGB 0-255 to Float 0.0-1.0  
  
    // 2. Inference (Offline)  
    output_array = New FloatArray[10] // Array to hold probabilities for 10 classes  
    ModelInterpreter.run(input_tensor, output_array)  
  
    // 3. Post-processing  
    max_score = 0.0  
    detected_label = ""  
  
    FOR i FROM 0 TO 9:  
        IF output_array[i] > max_score:  
            max_score = output_array[i]  
            detected_label = LABELS[i] // e.g., "Black Soil", "Red Soil"  
  
    // 4. Threshold Validation
```

```

IF max_score < CONFIDENCE_THRESHOLD (0.6):
    RETURN "Uncertain - Please Retake Photo"
ELSE:
    RETURN detected_label
END FUNCTION

```

Algorithm 2: Generative Advisory Synthesis

```

FUNCTION GenerateAdvisory(soil_type, user_notes):
    // 1. Fetch Contextual Data (Parallel Execution)
    gps_location = GetCurrentLocation()
    weather_data = OpenMeteoAPI.fetch(gps_location) // Temp, Humidity
    chem_data = ISRIC_API.fetch(gps_location)      // N, P, K, pH

    // 2. Construct Persona-Based Prompt
    system_role = "Act as a Senior Scientist at ICAR (Indian Council of Agricultural
    Research)."

    user_prompt = CONCATENATE(
        "Detected Soil: ", soil_type,
        "Nitrogen Level: ", chem_data.N,
        "Current Weather: ", weather_data.temp,
        "Farmer Observation: ", user_notes,
        "Task: Recommend 3 crops and calculating Fertilizer dosage in kg/acre."
    )

    // 3. API Call to Groq Cloud
    response = HTTP_POST(
        URL = "https://api.groq.com/openai/v1/chat/completions",
        MODEL = "llama-3.3-70b-versatile",
        BODY = {system_role, user_prompt}
    )

    // 4. Output Handling
    markdown_report = ParseJSON(response)
    RenderMarkdown(markdown_report)

```

```
TextToSpeech.speak(markdown_report)
END FUNCTION
```

Algorithm 3: The "Pivot Unit" Conversion Strategy

```
// Conversion Factors relative to 1 Acre
CONST FACTOR_ACRE = 1.0
CONST FACTOR_HECTARE = 2.47
CONST FACTOR_BIGHA = 0.4
CONST FACTOR_GUNTHA = 0.025

FUNCTION ConvertArea(input_value, from_unit, to_unit):
    // Step 1: Normalize Input to the Pivot Unit (Acre)
    // Example: Converting 2 Hectares -> 4.94 Acres
    value_in_acres = input_value * GET_FACTOR(from_unit)

    // Step 2: Convert from Pivot to Target Unit
    // Example: Converting 4.94 Acres -> Gunthas
    final_result = value_in_acres / GET_FACTOR(to_unit)

    RETURN Round(final_result, 4) // Return with 4 decimal precision
END FUNCTION
```

Algorithm 4: Secure Document Sandboxing

```
FUNCTION SecureFileStorage(source_uri):
    // 1. Prepare Secure Destination
    file_name = GetFileName(source_uri)
    internal_storage = App.getFilesDir() + "/secure_docs/"
    dest_file = CreateFile(internal_storage, file_name)

    // 2. Stream Copy (Byte-level transfer)
    input_stream = ContentResolver.open(source_uri)
    output_stream = FileOutputStream(dest_file)
```

```

WHILE (byte = input_stream.read()) != EOF:
    output_stream.write(byte)

CloseStreams()

// 3. Indexing in Local Database
DB.DocumentsTable.Insert(
    name = file_name,
    path = dest_file.getAbsolutePath(), // Private path
    timestamp = CurrentTime()
)

RETURN "File Locked Securely"
END FUNCTION

```

Algorithm 5: Geospatial Data Triangulation

```

FUNCTION FetchEnviroData(latitude, longitude):
    // 1. Define Asynchronous Tasks
    Task weather_task = ASYNC_CALL(
        OpenMeteoAPI.get(lat, lon, params="temp,humidity,moisture")
    )

    Task soil_chem_task = ASYNC_CALL(
        ISRIC_SoilGridsAPI.get(lat, lon, layers="nitrogen,ph,soc")
    )

    // 2. Parallel Execution & Synchronization
    // The app waits here until BOTH tasks complete or timeout
    TRY:
        Results results = AWAIT_ALL(weather_task, soil_chem_task)
    CATCH TimeoutException:
        RETURN "Offline Mode - Using Cached Data"

    // 3. Data Normalization & Fusion
    weather_obj = results.weather_task.body()
    soil_obj = results.soil_chem_task.body()

    // Convert raw API units to farming standards

```

```
final_data = {  
    temp_c: weather_obj.temperature_2m,  
    moisture_index: weather_obj.soil_moisture_index,  
    nitrogen_kg_ha: soil_obj.nitrogen_mean * 2.0, // Conversion factor  
    ph_level: soil_obj.phh2o_mean / 10.0      // pH is stored as int  
}  
  
    RETURN final_data  
END FUNCTION
```